# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** The sum of the first 12 terms of an AP is $510$ and its first term is $4$. The common difference is: 1

**(a)** $6$ **(b)** $8$ **(c)** $7$ **(d)** $9$

**2.** The exterior angle at $D$ of cyclic quadrilateral $ABCD$ (side $CD$ produced) equals the interior angle at $B$, which is $72^\circ$. The interior angle $\angle ADC$ is: 1

**(a)** $118^\circ$ **(b)** $98^\circ$ **(c)** $128^\circ$ **(d)** $108^\circ$

**3.** The area of a rhombus of side $13$ cm and one diagonal $10$ cm is (cm$^2$): 1

**(a)** 120 **(b)** 122 **(c)** 121 **(d)** 119

**4.** A bag has $36$ balls, red or blue. If $P(\text{red}) = \frac{5}{12}$, the number of blue balls is: 1

**(a)** 22 **(b)** 20 **(c)** 21 **(d)** 23

**5.** Three numbers in AP have sum $30$ and product $840$ (with $d > 0$). The largest number is: 1

**(a)** $18$ **(b)** $14$ **(c)** $15$ **(d)** $10$

**6.** $AB$ is a diameter, $C$ and $D$ are on the circle (same side of $AB$) and $ABCD$ is cyclic with $\angle DAB = 48^\circ$. Then $\angle BCD =$ 1

**(a)** $122^\circ$ **(b)** $142^\circ$ **(c)** $152^\circ$ **(d)** $132^\circ$

**7.** A triangle has sides $10$ cm, $17$ cm, $21$ cm. The length of the shortest altitude (cm) is: 1

**(a)** 8 **(b)** 10 **(c)** 7 **(d)** 9

**8.** A letter is chosen from the word MISSISSIPPI. The probability that it is not the letter I is: 1

**(a)** $\frac{7}{12}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{8}{11}$ **(d)** $\frac{7}{11}$

**9.** A town's population is 10000 and grows by 10% each year. After 2 years it is: 1

**(a)** $12000$ **(b)** $12100$ **(c)** $12300$ **(d)** $12200$

**10.** A chord $AB$ of length $14$ cm lies $24$ cm from the centre $O$. The area of triangle $OAB$ (cm$^2$) is: 1

**(a)** 170 **(b)** 167 **(c)** 169 **(d)** 168

**11.** A rope of length $88$ cm (take $\pi = \frac{22}{7}$) forms a circle. The same rope forms a square. The area of the square (cm$^2$) is: 1

**(a)** 484 **(b)** 485 **(c)** 483 **(d)** 486

**12.** The probability that a seed germinates is $\frac{7}{10}$. Out of $2000$ seeds the expected number that will NOT germinate is: 1

**(a)** 602 **(b)** 599 **(c)** 601 **(d)** 600

**13.** **Assertion (A):** The sequence $2, 5, 8, 11, \ldots$ is an arithmetic progression. 1
**Reason (R):** The difference between consecutive terms is the constant 3.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**14.** **Assertion (A):** Equal chords of a circle are equidistant from the centre. 1
**Reason (R):** The perpendicular from the centre of a circle to a chord bisects the chord.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**15.** **Assertion (A):** The circumference of a circle of radius 7 cm is 44 cm (take $\pi = \frac{22}{7}$). 1
**Reason (R):** The area of a circle of radius $r$ is $\pi r^2$.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** A circle of radius 7 cm is inscribed in a square (take $\pi = \frac{22}{7}$). Find the area of the square left uncovered by the circle. 2

**17.** A bag has 12 red balls and some blue balls. If $P(\text{blue}) = \frac{1}{4}$, find the number of blue balls. 2

**18.** Which term of the AP $5, 11, 17, \ldots$ is the first term to exceed $200$? 2

**19.** $ABCD$ is a cyclic quadrilateral and $\angle ABD = 32^\circ$, $\angle DBC = 41^\circ$. Find $\angle ABC$ and $\angle ADC$. 2

**20.** The area of a rhombus is $120$ cm$^2$ and one diagonal is $10$ cm. Find the other diagonal and the side of the rhombus. 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Find the sum of all natural numbers between 10 and 200 that leave remainder 2 when divided by 5. 3

**22.** $AB$ is a diameter of a circle; $C$, $D$ lie on the circle on the same side of $AB$ so that $ABCD$ is a cyclic quadrilateral with $\angle DAB = 34^\circ$. Find $\angle BCD$, $\angle ADB$ and $\angle ABD$. 3

**23.** The sides of a triangle are $13$ cm, $14$ cm and $15$ cm. Find its area and the altitudes on the sides $13$ cm and $14$ cm. 3

**24.** A letter is chosen at random from the word MISSISSIPPI. Find the probability that it is 3
(i) not S,
(ii) a vowel or S.

**OR**

Three positive numbers are in GP. Their product is $216$ and the sum of the first and third is $15$. Find the numbers. 3

**25.** Two circles of radii $13$ cm and $20$ cm intersect in a common chord of length $24$ cm. Find the distance between their centres if they are on 3
(i) opposite sides,
(ii) the same side of the common chord.

**OR**

A rope $88$ cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26. Case study.** A circular pond of radius 14 m in a village in Kerala is surrounded by a 7 m wide path. Take $\pi = \dfrac{22}{7}$. 5
(i) Find the circumference of the pond.
(ii) Find the area of the pond.
(iii) Find the radius and the area of the outer circle formed by the path.
(iv) Find the area of the path and the cost of paving it at Rs 40 per $m^2$.

**27.** A bag has 7 red, 8 blue and 12 green balls. Some green balls are removed so that $P(\text{green})$ becomes $\frac{1}{4}$. How many green balls were removed? 5

**OR**

A ball is dropped from 81 m and rebounds to 2/3 of the previous height. Show that the rebound heights form a GP and find the 3rd rebound height. 5

## END OF QUESTION PAPER — Total: 50 Marks

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